$$\Delta x.(m.\Delta v) \geq \frac{h}{4\pi}$$

$$\Delta v \geq \frac{h}{4\pi.\, m\, .\, \Delta x}$$

$$\Delta v \geq \frac{6.626 \times 10^{-34}\ kg\ m^2 s^{-1}}{4 \times 3.14 \times 9.11 \times 10^{-31}\ kg \times 2.645 \times 10^{-13}\ m}$$

$$\Delta v \geq 2.189 \times 10^{8}\ ms^{-1}$$

Therefore, the uncertainty in the velocity of the electron is comparable with the velocity of light. At this high level of uncertainty it is very difficult to find out the exact velocity.

Evaluate Yourself

2. Calculate the uncertainty in the position of an electron, if the uncertainty in its velocity is $5.7 \times 10^5\ ms^{-1}$.

2.4 Quantum mechanical model of atom – Schrödinger Equation:

The motion of objects that we come across in our daily life can be well described using classical mechanics which is based on the Newton's laws of motion. In classical mechanics the physical state of the particle is defined by its position and momentum. If we know both these properties, we can predict the future state of the system based on the force acting on it using classical mechanics. However, according to Heisenberg's uncertainty principle both these properties cannot be measured simultaneously with absolute accuracy for a microscopic particle such as an electron. The classical mechanics does not consider the dual nature of the matter which is significant for microscopic particles. As a consequence, it fails to explain the motion of microscopic particles. Based on the Heisenberg's principle and the dual nature of the microscopic particles, a new mechanics called quantum mechanics was developed.

Erwin Schrödinger expressed the wave nature of electron in terms of a differential equation. This equation determines the change of wave function in space depending on the field of force in which the electron moves. The time independent Schrödinger equation can be expressed as,

$$\hat{H}\Psi = E\Psi \text{ -------(2.12)}$$

Where $\hat{H}$ is called Hamiltonian operator, Ψ is the wave function and is a funciton of position co-ordinates of the particle and is denoted as $\Psi(x,y,z)$ E is the energy of the system

$$\hat{H} = \left[\frac{-h^2}{8\pi^2 m}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right)+V\right]$$

(2.12) can be written as,

$$\left[\frac{-h^2}{8\pi^2 m}\left(\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}\right)+V\psi\right] = E\psi$$

Multiply by $-\frac{8\pi^2 m}{h^2}$ and rearranging

$$\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}+\frac{8\pi^2 m}{h^2}(E-V)\psi = 0$$

------(2.13)

The above schrödinger wave equation does not contain time as a variable and is referred to as time independent Schrödinger wave equation. This equation can be solved only for certain values of E, the total energy. i.e. the energy of the system is quantised. The permitted total energy values are called eigen values and corresponding wave functions represent the atomic orbitals.

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